Investigating the Association Between High School Graduation Rate and Proportion of Annual Crimes by American State Through Linear Regression T-Test

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AGENDA

PROJECT BACKGROUND

We chose this topic because we wanted to investigate how education (or the lack thereof) affects the proportion of annual crimes in each state in America.

RESEARCH QUESTION:

Is there a relationship between a state's high school graduation rate and its proportion of annual crimes?

PARAMETER

Parameter: The slope of the best-fit line between state high school graduation rate and proportion of crime rate (B).

HYPOTHESES

Ho: There is no linear relationship between a state's high school graduation rate and its proportion of crime rate (B = 0)

Ha: There is a linear relationship between a state's high school graduation rate and its proportion of crime rate (B ≠ 0)

DATA COLLECTION

We used the graduation rates posted by the states' Departments of Education to collect data on the graduation rates of all US states (including Washington D.C.) and the World Population Review to collect data on the number of total crimes committed annually in each state. We then compiled the data points into an excel table and graphed using the Insert Graph function.

ABOUT OUR DATA SOURCES

U.S. DEPARTMENT OF EDUCATION

- **PURPOSE:** We are using this source to find the high school graduation rate per state, specifically with the average Adjusted Cohort Graduation Rate (ACGR) for public high schools in each state. This source provided data from the school year of 2019-20.
- **CREDIBILITY:** We believe this source is credible because it is published from an official U.S. government webpage.

ABOUT OUR DATA SOURCES

WORLD POPULATION REVIEW

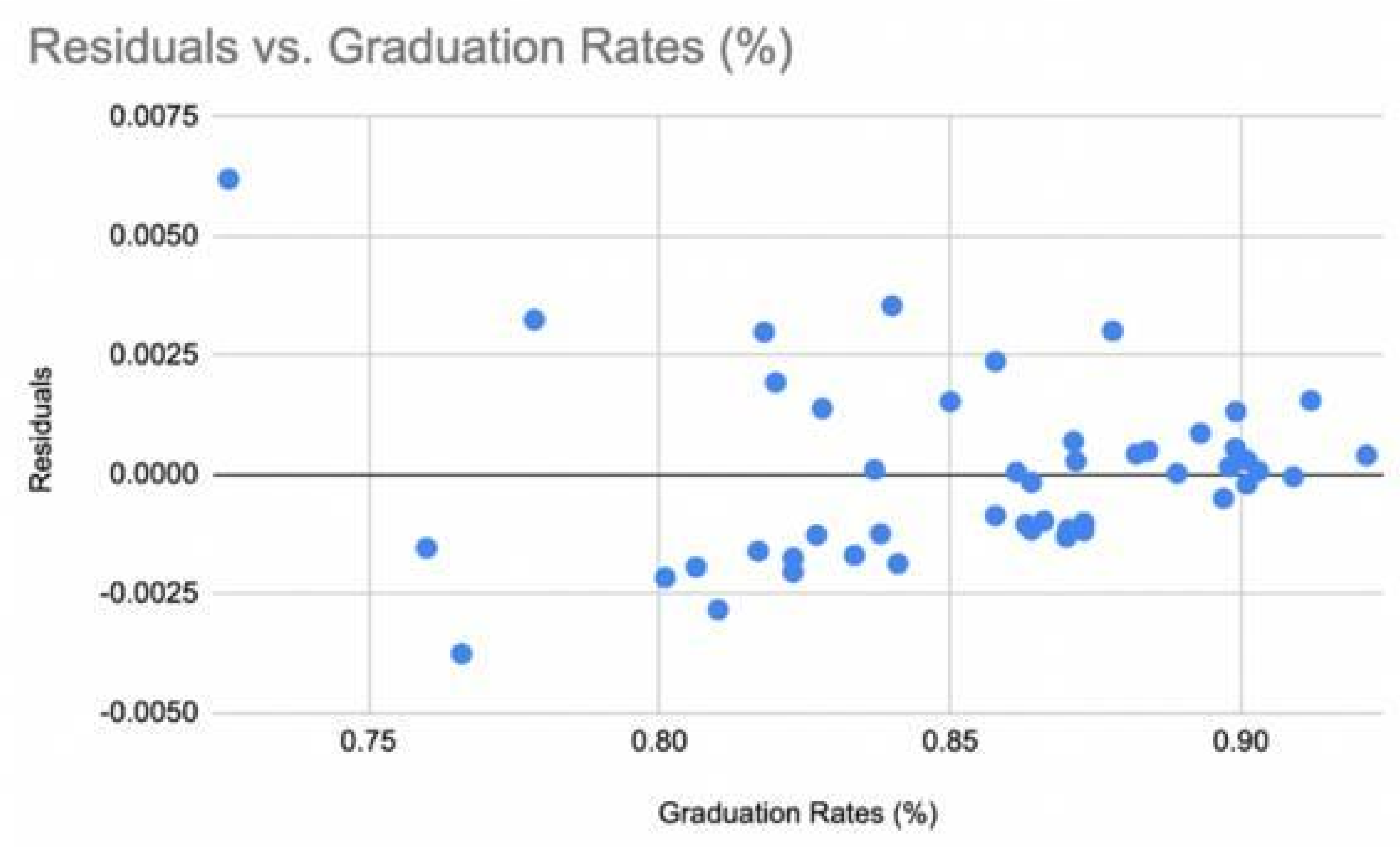
- **PURPOSE**
    - We are using this source to find the crime rate per state, specifically with FBI information regarding variables that affect crime and released data found in their Crime Data Explorer.
- **CREDIBILITY**
    - We believe that this source is credible because it pulls from many other reputable sources such as the FBI, and then compiles all the data to give an accurate view of all state crimes.

OUR DATA

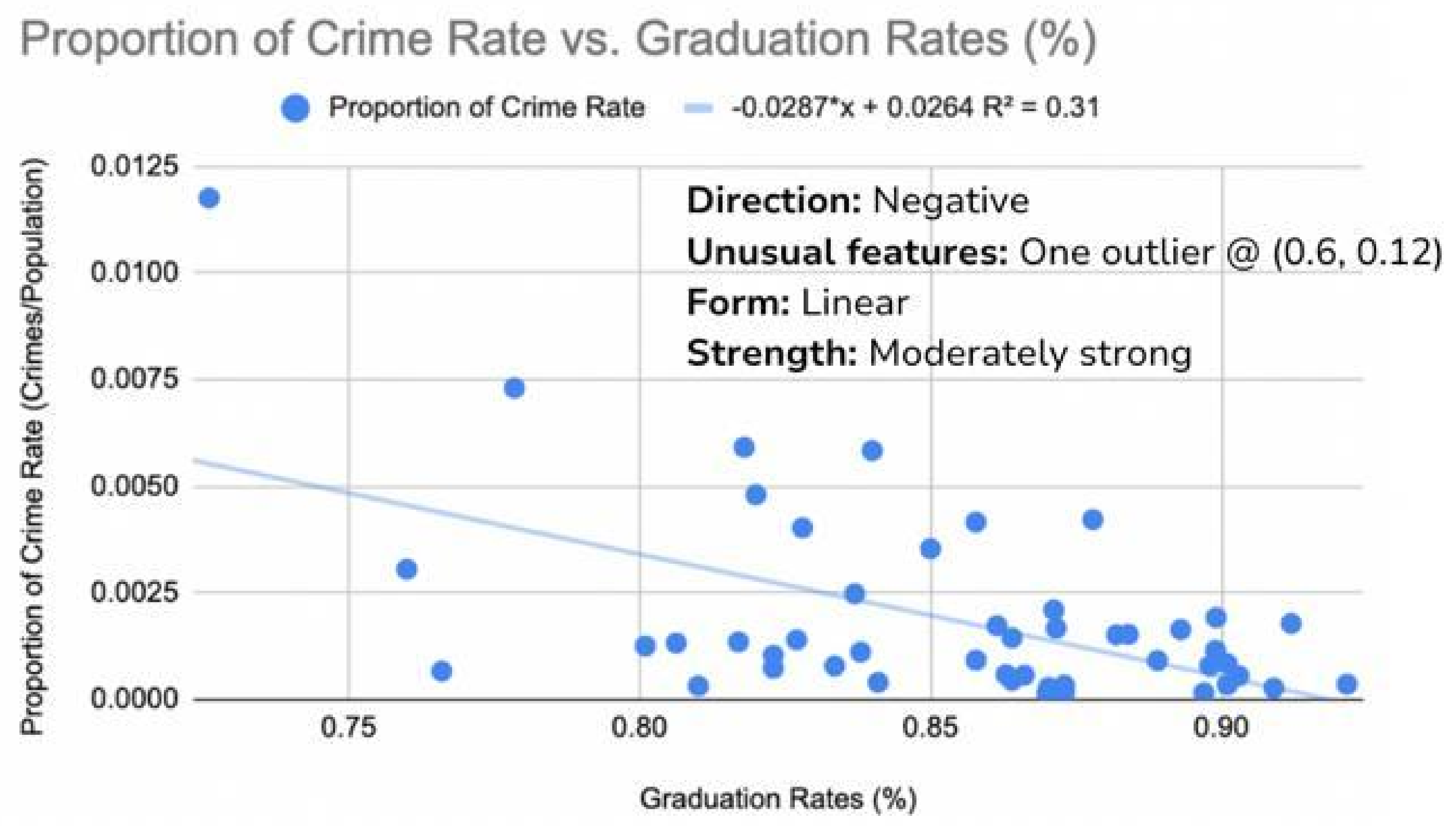
	A	B	C	D	E
1	**State**	**Graduation Rates**	**Population**	**Crime Rate**	**Proportion of Crime Rate**
2	DC	0.726	678972	7986	0.01176189887
3	New Mexico	0.76	2114371	6462	0.00305622807
4	Louisiana	0.827	4573749	6408	0.001401038841
5	Colorado	0.823	5877610	6091	0.001036305573
6	South Carolina	0.838	5373555	5973	0.001111554641
7	Arkansas	0.8991	3067732	5899	0.001922918951
8	Oklahoma	0.864	4053824	5870	0.001448015503
9	Washington	0.823	7812880	5759	0.0007371161467

GRAPHS OF DATA

Graph of Residuals



Graph of Data



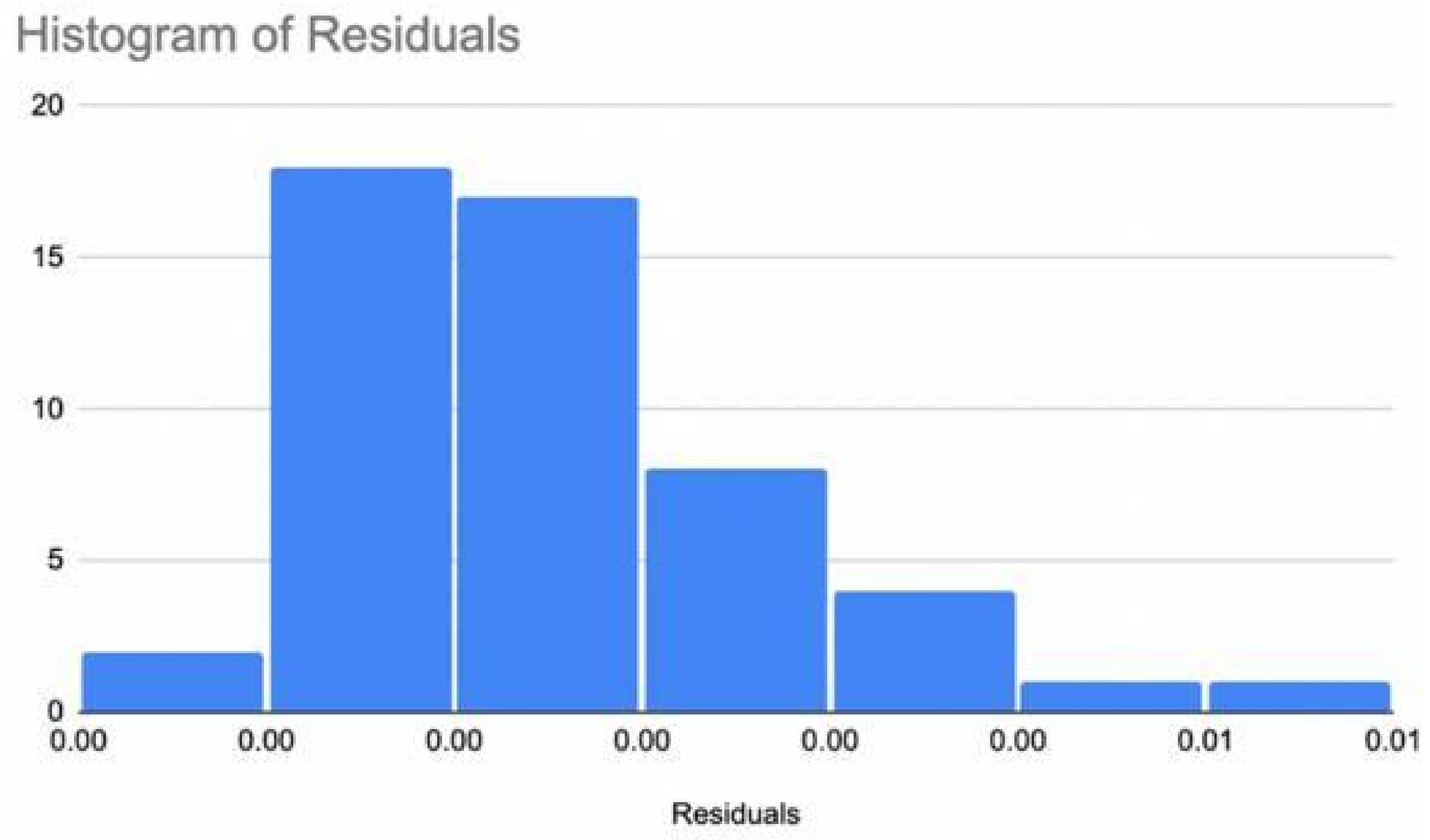
DATA ANALYSIS

- **To test our investigate question, we performed a Linear Regression T-Test for Slope to examine the association between a state's high school graduation rate and proportion of crime rate.**
- **Conditions for Inference:**
  - Random:
    - NO, but it is a representative sample. Proceed with caution
  - Independent:
    - YES, one state's result in crime rates does not affect the data collected from other states.

DATA ANALYSIS

- **Conditions for Inference (cont'd.)**
  - Equal variance:
    - YES, but graph of residuals shows slight pattern - seems alright but proceed with caution
  - Normal:
    - YES, sample size of 51 > 30 (CLT)
  - Linear:
    - YES, looks fairly linear - another model might be a better fit, but this represents as well

Histogram of residuals



RESULTS OF LINEAR REGRESSION

3. Enter data

Help me arrange the data

Label:	State High School G	State's Proportion of C
Values:	0.726	0.0001734574998
	0.76	0.0008435417796
	0.827	0.001153724785
	0.823	0.000283759447
	0.838	0.0005855747329
	0.8991	0.0002824771196
	0.864	0.00591551852
	0.823	0.0009156314208
	0.898	0.0005574395339
	0.8063	0.0003188120743
	0.8997	0.0001627389938
	0.7785	0.001782414378
	0.882	0.000358089507
	0.85	0.002483662755
	0.766	0.004026478692
	0.897	0.0002703738015
	0.864	0.001255136849
	0.893	0.001734586114
	0.8578	0.0003449310631
	0.87	0.001671832897

P value and statistical significance:

The two-tailed P value is less than 0.0001

By conventional criteria, this difference is considered to be extremely statistically significant.

Confidence interval:

The mean of State High School Graduation Rate minus State's Proportion of Crime Rate Over Population

0.853390305694198

95% confidence interval of this difference: From 0.841602461215339 to 0.865178150173058

Intermediate values used in calculations:

t = 143.6311

df = 100

standard error of difference = 0.006

Review your data:

Group	State High School	State's Proportion of
Mean	Graduation Rate	Crime Rate Over
SD	0.855214705882353	Population
SEM	0.042374877338014	0.001824400188155
N	0.005933669679955	0.002183328018121
	51	0.000305727074068
		51

CONCLUSION

Because our p-value is <0.0001, lower than common statistical significance levels of 0.05 and 0.01, we reject our null hypothesis (Ho). We have convincing evidence of a linear relationship between a state's high school graduation rate and its proportion of crime with respect to its population. We still acknowledge that we were only able to make this conclusion after proceeding with caution because not all of our conditions were met, so we take this into consideration.

THANK YOU!